

Michael Jackson

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Education

B.S. Computer Engineering, University of California, Santa Cruz 3.60 GPA

Relevant Coursework: Advanced Multivariate Calculus, Linear Algebra, Computer Architecture, Data Structures and Algorithms, Programming Abstractions: Python, Computer Systems, Discrete Mathematics, Parallelism and Concurrency, Advanced Programming, Computer Networks, Network Security, Circuit Analysis

Experience

Marchup Internship(3 months): Contributed to efforts to horizontally scale a social media service at MarchUp by re-architecting a web-server-hosted search engine deployed on a monolithic PHP/Apache stack. Designed and evaluated multiple AWS-based distributed prototypes, including a real-time indexing architecture that improved search freshness and scalability.

Reference: Harsh Parandekar (Boss/Manager) Email(best): harsh@marchup.net Phone: 408-230-9028

Dymium Internship(8 months): Implemented a scalable Kubernetes architecture for a stateful reverse tunneling proxy using a custom DNS-based load-balancing protocol, enabling horizontal scaling of previously single-instance servers.

Reference: Igor Plotnikov(Boss/Manager) Email(best):igor@dymium.io Phone: 408-702-7865

QUIC Research University of California, Santa Cruz (6 months): Conducted research under Professor Ram Sundara Raman in network systems and security, developing a QUIC protocol fuzzer to evaluate firewall behavior. Designed packet mutation strategies to identify evasion and non-evasion cases, systematically documenting results, and co-authored a paper analyzing fuzzing results on open-source firewalls.

Reference: Ram Sundara Raman (Professor) Email(best): rsundar2@ucsc.edu

Projects

QUIC Fuzzer <https://github.com/r-andlab/mutate-quic>

A network security fuzzing framework designed to evaluate firewall behavior on QUIC traffic. The system generates and captures valid QUIC packets, applies protocol-aware mutation strategies, and replays modified traffic against firewall-protected targets, recording and analyzing evasion and non-evasion outcomes to characterize firewall handling of encrypted transport protocols. Findings were systematically analyzed and synthesized into written research artifacts, emphasizing critical evaluation of firewall behavior and protocol handling.

https://github.com/michaelkamensky/Fuzzing-Firewalls-with-QUIC/blob/main/cse195_senior_thesis.pdf

Custom DNS server

A scalable reverse tunneling system built on Kubernetes that replaces single-instance proxies with a horizontally scalable design. The architecture employs a custom DNS server deployed as a Kubernetes Service and propagated to workloads via ConfigMaps, enabling dynamic traffic routing to active tunneling proxies through a DNS-based load-balancing protocol using zero-TTL records and proxy-reported state via API calls.

Skills

Primary Focus (Production & Research Experience)

C (4+ years) - systems programming, networking, embedded systems

Python (6+ years) - scripting, analysis, tooling

Go (1+ year) - distributed systems, networking tools

Networking & Security

QUIC protocol analysis (6 months)

Protocol fuzzing & firewall evaluation (6 months)

Network security & encrypted transport analysis (coursework + research)

Distributed Systems & Infrastructure

Kubernetes (8 months) - stateful services, DNS-based routing, scaling

AWS (3 months) - distributed prototypes, backend services

DNS-based load balancing (project experience)

Additional Languages & Technologies

Java (coursework + projects)

C++ (coursework)

RISC-V Assembly (coursework + labs)

PHP (project experience)

HTML/CSS/JavaScript (supporting knowledge)